



Uganda National Roads Authority

DIRECTORATE OF NETWORK PLANNING & ENGINEERING (UNRA)

A PRESENTATION TO THE BOARD OF DIRECTORS ON ROAD NETWORK ASSESSMENT AND PLANNING



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Manager Network Planning
November 2020

1. Introduction
2. Data Collection
3. Data Management
4. Optimization Process
5. Challenges

Introduction: Mandate and Staffing

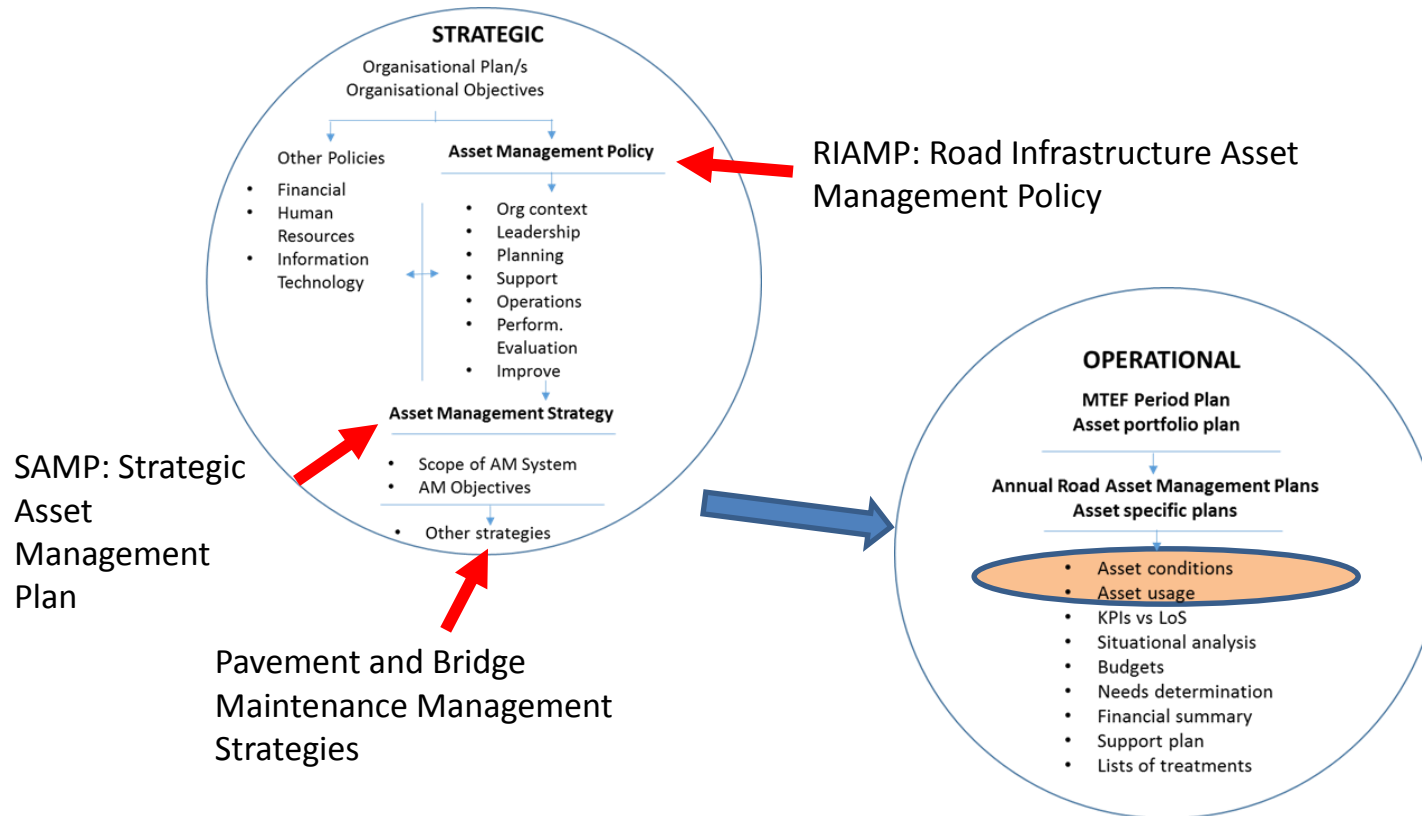
- Collection and updating of road network and bridge asset data;
- Maintenance of Road, Bridge data bank and Road Management Systems;
- Formulate short and long term bridge and road investment plans;
- Collect road condition and traffic data to support development and maintenance programming of the network.

Position		Number of Approved Structure	Number of Staff Recruited
Head, Network planning	Network	1	0
Manager Network planning	Network	1	1
Network Engineers		2	1
Network Technicians		5	5
Graduate Engineers		2	2
TOTAL		11	9

82%

- ❖ *Asset Management Skill sets are required*
- ❖ *High competence in specialized IT packages both for collection and analysis*

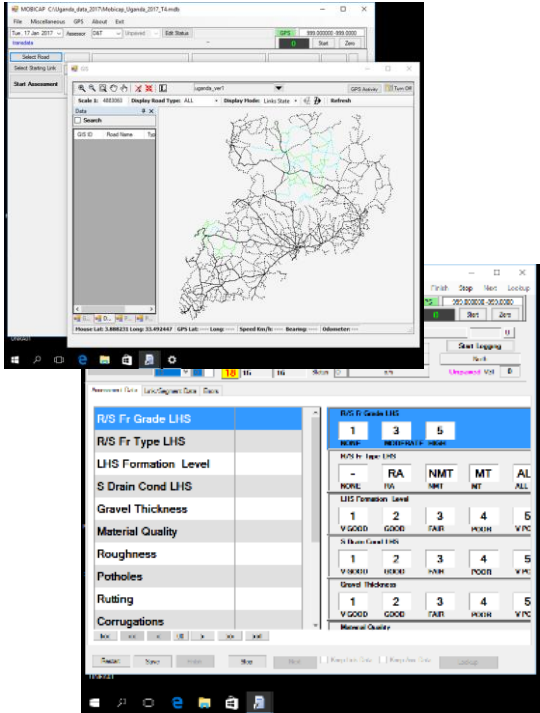
Asset Management Framework

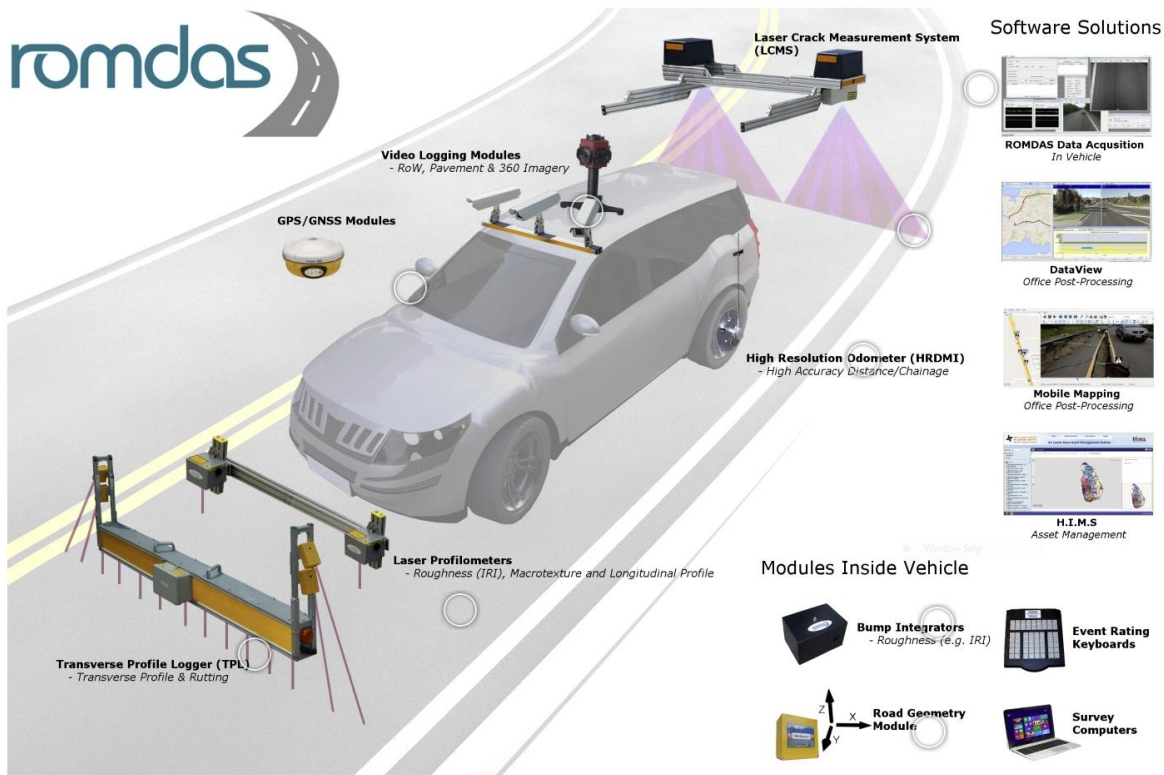


- Draft Data Collection Policies and Procedures
- Draft Quality Management Inspection for Visual and Inventory of Paved and Unpaved
- Draft Roads Guidelines For Road Panel Inspections
- Draft Guidelines For Road Performance Indicators
- Draft Manual Visual inspections manual of paved and unpaved roads Feb 2017
- Road Infrastructure Asset Management Policy
- Draft Pavement & Bridge Maintenance Management Strategy
- PMS User Manual Dec 2017
- RMS User Manual Dec 2017
- Bridge Management System (BMS) User Manual

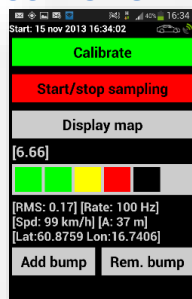
- **Inventory:** Road, Link, Bridge, Culverts, Point Features and Line Features (19No –road signs, place marks, etc.), Location, alignment, Type, Age, Specification, etc
- **Visual condition data:** 35No attributes (edge breaks, raveling, corrugations, width loss, cracks, pot-holes, etc.
- **Equipment based data:** Roughness, Rutting, Video, GPS centerline, Deflection, GPR, etc.
- **Traffic:** Manual Classified Count and Automated traffic counts, Origin destination, crowd sourced OD.
- **Axle load**

Equipment Used





Rapid Condition assessment - Roadroid



Calibration and Assessment

- Visual assessment data depends on the
 - knowledge,
 - experience and
 - commitment of the assessors.
- Annual training and calibration sessions

Procedure For Assessment

1. In cooler months, end of rainy season, under dry conditions only and complete within 1-2 months
2. Not more than 100 km/day @ 25 km/h
3. Minimum of 2 stops per segment (at start, end, also where major changes occur)
4. Assess the full paved width, concentrate on the slow lane on multi-lane roads
5. Be aware of the sun's position!
6. **ONLY RATE WHAT WE SEE!**

Inventory

Functional Classification	Bituminous , km	Unsealed, km	Total, km	% Total	Description
A	2594	0	2594	12.3%	International Trunk Roads
B	1395	1447	2842	13.5%	National Trunk Roads
C	1323	14165	15489	73.7%	Primary Roads
M	85	0	85	0.4%	Expressways
Total	5398	15612	21010		

Bridges	379No
Major Culvert Structures	293No

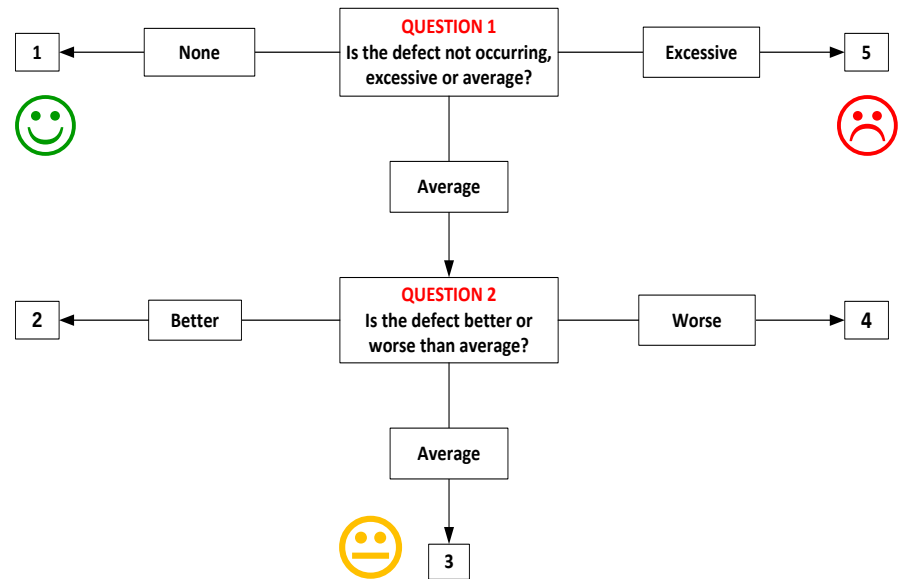
Manual Classified Count	297No
Automated Traffic Count	15No

- Roads and Road links (unique link number)
 - A, B and C roads
 - Eg: A001_Link08; B304_Link03; C356_Link02
- Bridges B001, Culverts C787
- TCS 001,002....
- Attributes 19No and 25No

Condition Rating System

Paved Defects (8)

Roughness
'All' Cracking
'Wide' Cracking
Potholes
Width Loss
Edge Drop
Surface Deterioration
Rutting



Unpaved Defect (9)

Gravel Thickness
Roughness
Potholes
Rutting
Corrugations
Erosion Gullies
Drainage Condition
Drainage Formation Level
Roadside Friction (Type and Grade)

- Segment = 1km
- Main characteristics of distress
 - Type and Severity (grade) of occurrence



Condition Defects Rated

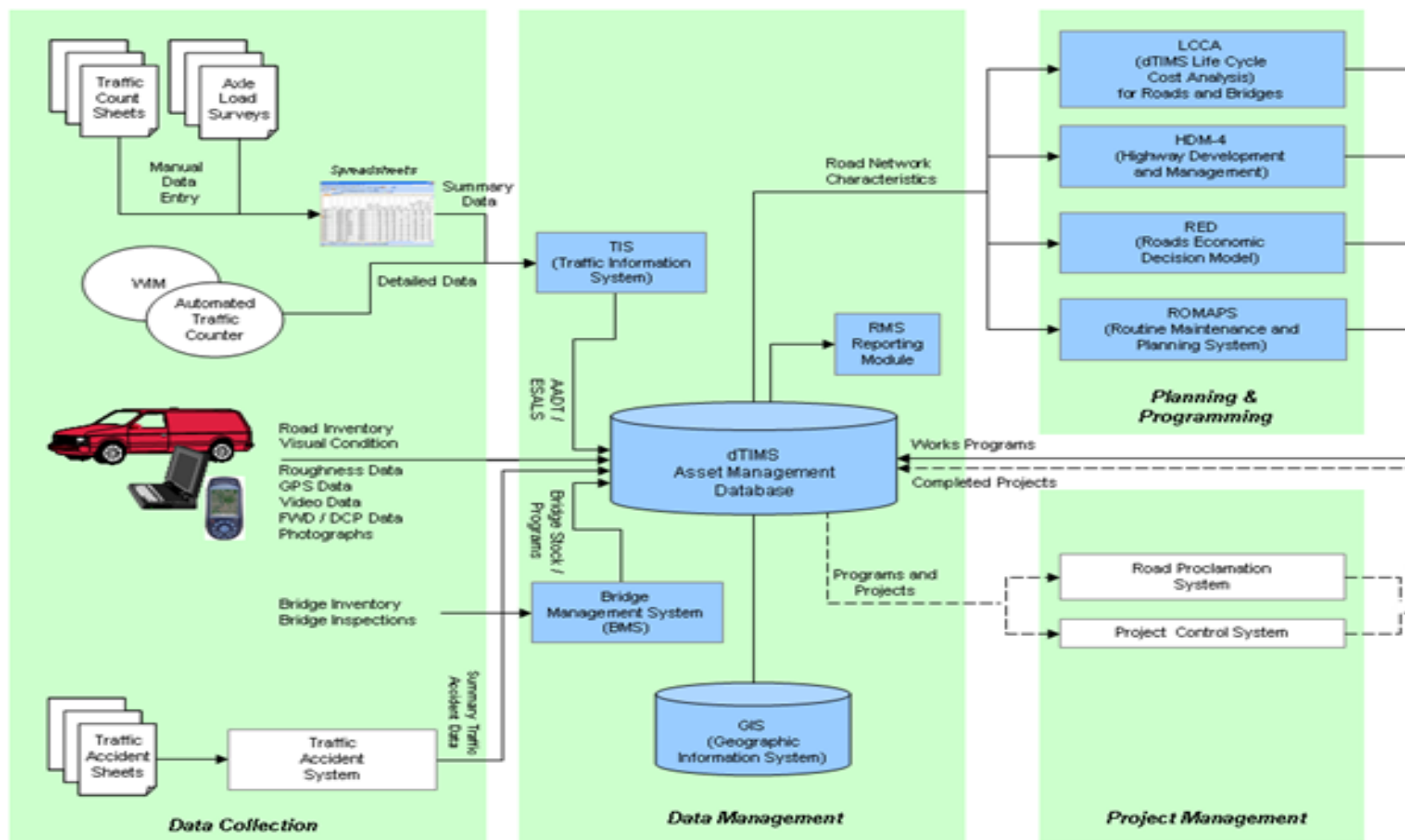
ALL CRACKS-Paved

Grade	Description
1	No cracking.
2	Isolated or continuous all cracking; Area of all cracking < 5% of surface area of segment.
3	Isolated or continuous all cracking; Area of all cracking $\geq 5\%$ and $< 15\%$ of surface area of segment.
4	Isolated or continuous all cracking; Area of all cracking $\geq 15\%$ and $< 30\%$ of surface area of segment.
5	Isolated or continuous all cracking; Area of all cracking $\geq 30\%$ of surface area of segment.

ROUGHNESS - Unpaved

Grade	Description
1	Ride very smooth and very comfortable; No/slight unevenness of the profile; No rutting or potholes; IRI value < 5 .
2	Ride smooth/fair and comfortable; Moderate unevenness of the profile; Moderate rutting but no potholes; IRI value ≥ 5 and < 7 .
3	Ride poor and uncomfortable; Frequent unevenness of the profile Significant rutting, corrugations and occasional potholes; Moderate speed reduction; IRI value ≥ 7 and < 9 .
4	Ride very poor and very uncomfortable; Severe unevenness of the profile; Extensive rutting, corrugations and several potholes; Driving speed much lower than speed limit; Road unsafe owing severe unevenness; IRI value ≥ 9 and < 16 .
5	Impassable except by 4 Wheel Drive vehicles. IRI value ≥ 16 .

Data Management - The RMS





Condition Calculation Parameters

VCI/RCI Weights - Paved

Defect	Weighting (%)	Range Considered
All Cracks	7	0 – 40%
Wide Cracks	10	0 – 30%
Ravelling & Potholes	12.5	0 – 100 potholes
Rutting (visual or measured)	10	1-5
Bleeding	2.5	1-5
Roughness (measured)	20	1 – 8
Drainage	2.5	1-5
Edgedrop & Widthloss	10	1-5
Shoulder condition	3	1-5
Side drain condition	2.5	1-5
FWD (SNP number calculated from measured FWD)	15	2-7
Weight on Carriage way	94.5	
Weight on Shoulder	5.5	

VCI/RCI Weights - Unpaved

Defect	Weighting (%)
Drainage Formation (L & R)	16
Drainage Condition (L & R)	14
Gravel Thickness	20
Material Quality	17
Roughness	10
Potholes	5
Rutting	5
Corrugation	5
Erosion Gullies	8

Example - Unpaved

Defect	Weighting (%)	Dist. Value per band	1	2	3	4	5
Drainage Formation (L & R)	16	4	16	12	8	4	0
Drainage Condition (L & R)	14	3.5	14	10.5	7	3.5	0
Gravel Thickness	20	5	20	15	10	5	0
Material Quality	17	4.25	17	12.75	8.5	4.25	0
Roughness	10	2.5	10	7.5	5	2.5	0
Potholes	5	1.25	5	3.75	2.5	1.25	0
Rutting	5	1.25	5	3.75	2.5	1.25	0
Corrugation	5	1.25	5	3.75	2.5	1.25	0
Erosion Gullies	8	2	8	6	4	2	0

=VALUE(IF(S13=5,"0",IF(S13=4,"4",IF(S13=3,"8",IF(S13=2,"12",IF(S13=1,"16"))))))



Example - Unpaved

Link	Link Start	Start	End	Road Type	Road width	LHS Formation Level	Sdrain Cond LHS	Gravel Thickness	Material Quality	Roughness	Potholes	Rutting	Corrugations	Erosion/Gullies	Sdrain Cond RHS	RHS Formation Level
C67501	0	0	1	G	10.7	1	1	1	1	2	2	3	2	1	1	1
C67501	0	1	2	G	10.5	1	1	1	1	2	2	2	2	1	1	1
C67501	0	2	3	G	10.5	1	1	1	1	2	2	3	3	1	1	1
C67501	0	3	4	G	10.5	1	1	1	1	3	2	3	2	3	1	1

Formation Level (AVG)	Sdrain Cond (AVG)	Formation Level	Sdrain Cond	Gravel Thickness	Material Quality	Roughness	Potholes	Rutting	Corrugations	Erosion/Gullies	VCI check	Segment length	Weighted VCI
1	1	16.00	14.00	20.00	17.00	7.50	3.75	2.50	3.75	8.00	92.5	1.0	92.50
1	1	16.00	14.00	20.00	17.00	7.50	3.75	3.75	3.75	8.00	93.75	1.0	93.75
1	1	16.00	14.00	20.00	17.00	7.50	3.75	2.50	2.50	8.00	91.25	1.0	91.25
1	1	16.00	14.00	20.00	17.00	5.00	3.75	2.50	3.75	4.00	86	1.0	86.00

[illegible][illegible]



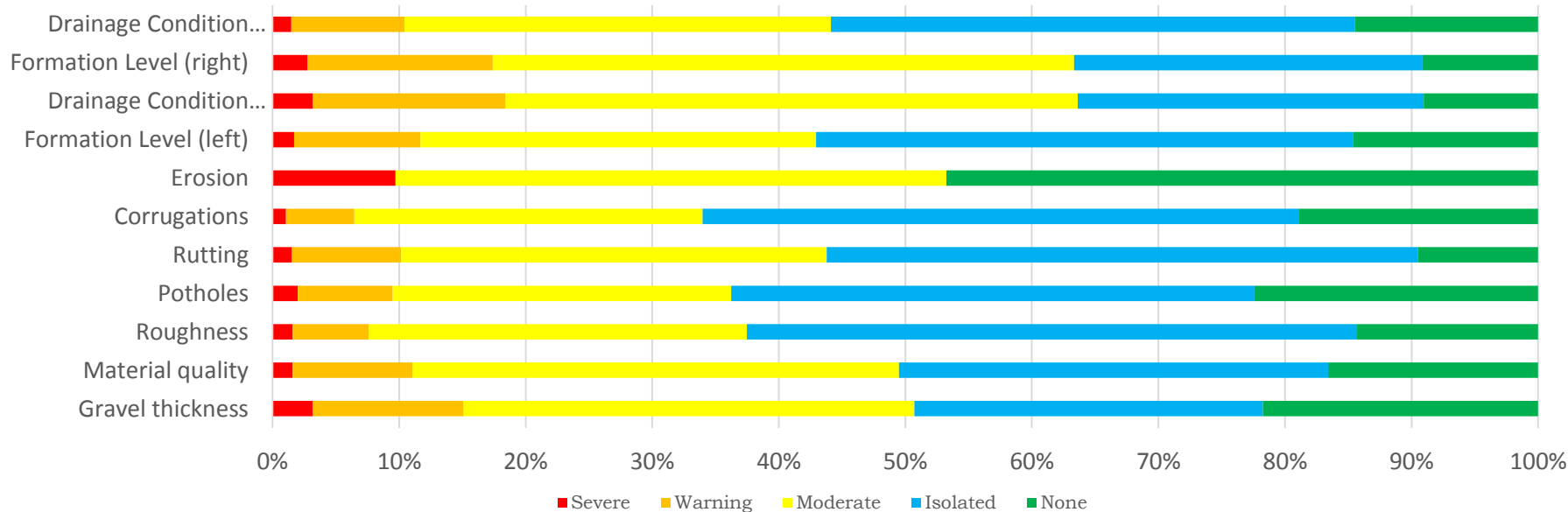
Example – Unpaved Results

VCI rating	VCI range	Bins	Frequency
Very Good	86% to 100%	100	1340
Good	71% to 85%	85	2860
Fair	51% to 70%	70	4375
Poor	31% to 50%	50	1934
Very Poor	0% to 30%	30	524

11033

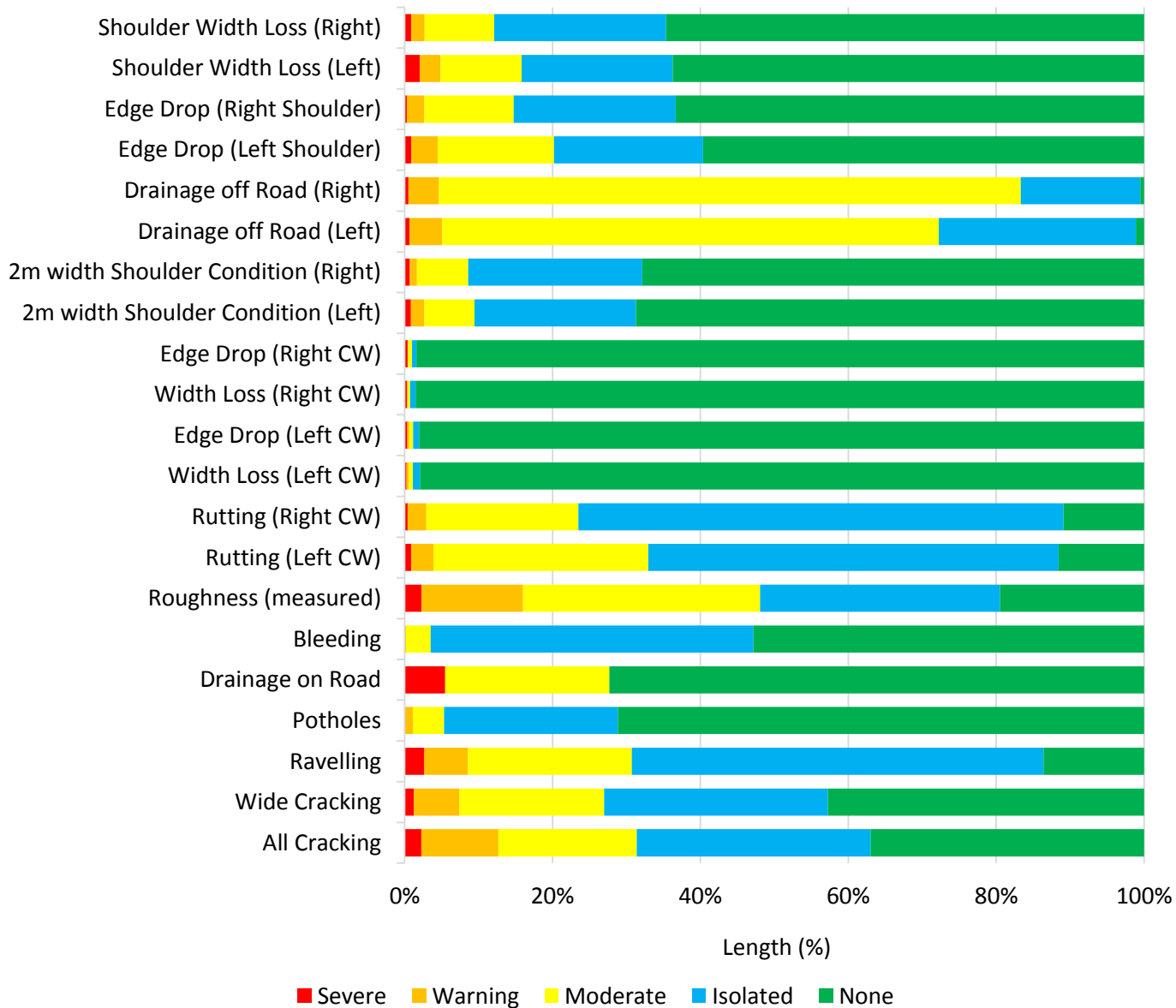
VCI rating	VCI range	Bins	% Frequency
Very Good	86% to 100%	100	12%
Good	71% to 85%	85	26%
Fair	51% to 70%	70	40%
Poor	31% to 50%	50	18%
Very Poor	0% to 30%	30	5%

100%



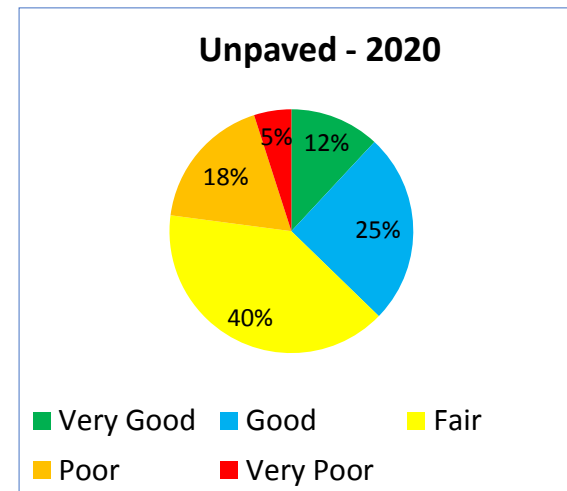
Example – Paved Results

Paved roads 2020

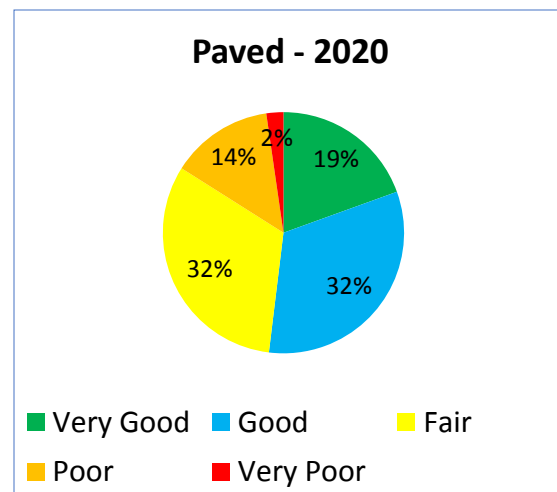


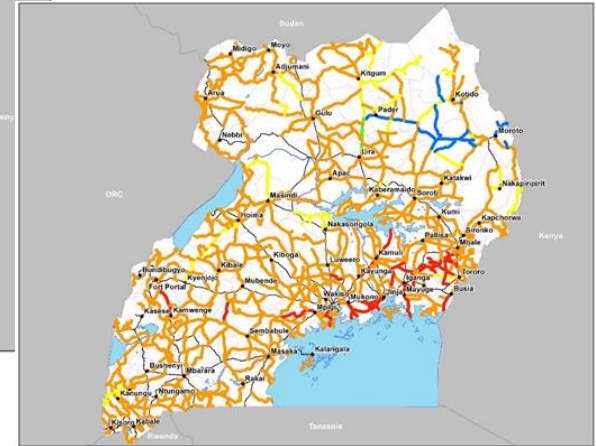
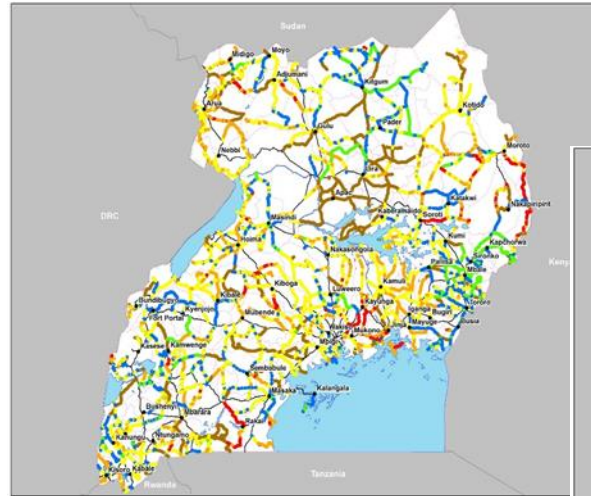
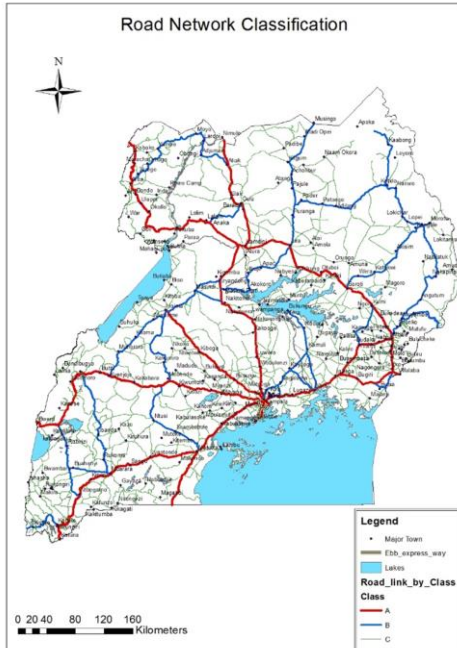
Summary Condition

VCI Rating for unpaved roads	Percentage	Length (km)
Very Good	12%	1263
Good	25%	2682
Fair	40%	4215
Poor	18%	1902
Very Poor	5%	524
Total		10586



Roughness Rating for paved roads	Percentage	Length (km)
Very Good	19%	549
Good	32%	914
Fair	32%	903
Poor	14%	387
Very Poor	2%	64
Total		2817





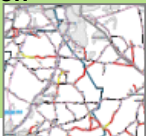
- Annual Road Condition Report - *Performance*
- Bi annual Automated Traffic Report – *Growth rates and Seasonal Variation factors*
- Bridge Condition
- Provisional acceptance testing reports
- Design inputs report (Axle load, pavement surveillance and traffic Reports)

LIFE CYCLE COST ANALYSIS

Input

Data

Network definition



Current and Historic Condition data

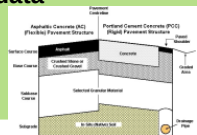
- Visual survey
- Profile data



Traffic data



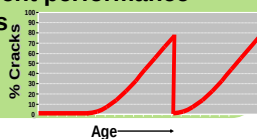
Pavement data



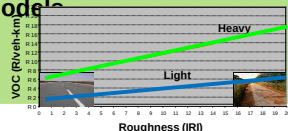
Models & Information

Pavement performance models

- HDM-4



Vehicle operating cost models



Maintenance

- Committed treatments
- Triggers
- Unit costs
- Policies



Budget Scenarios

- MTEF
- Any other fund level



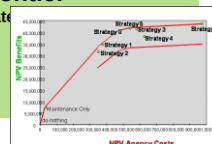
Life cycle cost analysis with dTIMS

Output

Decision support

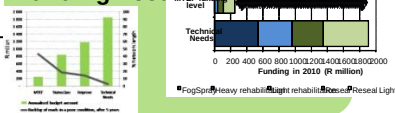
Efficiency frontier

- Generate strategies
- B vs C

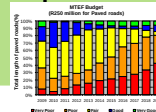


Support to planning and budgeting

Funding needs



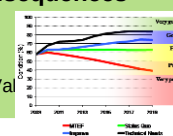
Determine effect of maintenance policies on road network



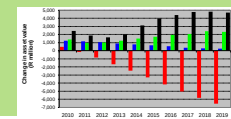
Reporting

Condition consequences

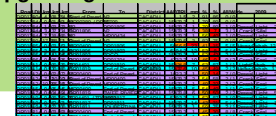
- Predicted CI
- Predicted VOC
- Predicted EUC
- Predicted Asset Value



Needs Report



Maintenance, Rehabilitation and Upgrading Plans



Geographic maps

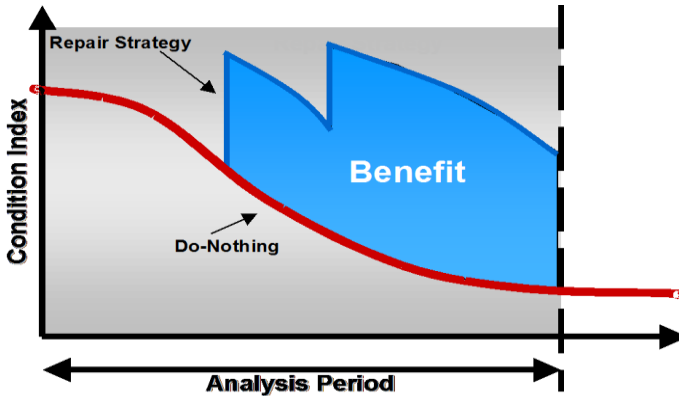
- MTEF
- Any other fund level

Treatments and Triggers

Treatment	Cost in US\$	Trigger
Reseal	10 / m ²	All cracks > 2% and wide cracks <= 15% and roughness <= 4.2 IRI and rutting <= 20mm and AADT >= 4000 or Wide cracks between 5% and 10% and AADT < 4000 and roughness <= 4.2 IRI and rutting <= 20mm
Reseal Light	8 / m ²	All cracks > 1% and wide cracks <= 5% and roughness <= 4.25 IRI and rutting <= 20mm and AADT < 4000
Light Rehabilitation	45 / m ²	Wide cracks > 10% and condition > 45% and roughness <= 6.6 IRI and rutting < 20mm or Wide cracks <= 10% and roughness > 5.5 but < 6.6 IRI and rutting < 20mm or Roughness <= 5.5 and wide cracks > 10% and rutting < 20mm
Heavy Rehabilitation	60 / m ²	Roughness > 5.5 IRI and wide cracks > 10% or Roughness > 6.6 IRI or Rutting > 20mm or Overall pavement condition very poor
Regravel	13,500 / km	The option to regravel the road is triggered when the wearing course thickness is less than 30mm.
Upgrade to standard paved road	1,050,000 / km	Traffic > 200 AADT and economic merit, i.e. savings in road user costs exceed the cost of the upgrade. Total transport costs (TTC) must be positive

Optimization Function

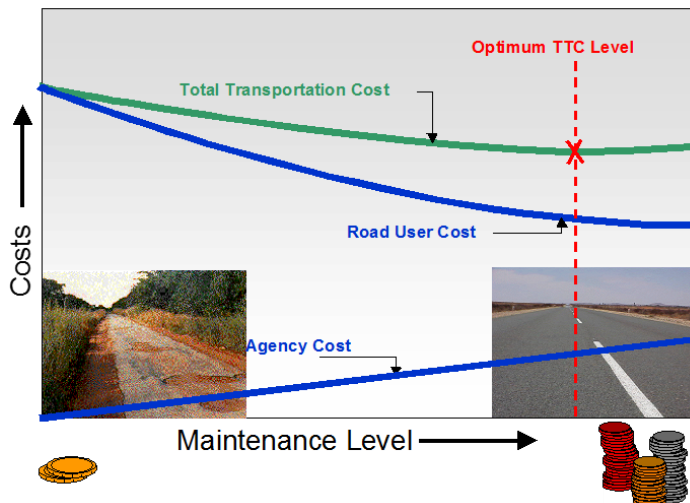
Area-Under-the-Condition Curve – Used for paved roads



$$\text{Present Value Benefit (i)} = \text{AADT (i)} * (\text{Str_Cond(i)} - \text{DN_Cond(i)})$$

Highest PV Benefits roads are triggered first

Total transport cost (TTC)



- TTC functions only used to determine which unpaved roads have economic merit for upgrading to paved standards
- Ensures the combined sum of the costs to the roads authority and road users are minimised

Funding Scenarios

Funding Scenario		Description
Constrained funding scenarios	"Received Budget or Status Quo" US\$ 1,000 million	Investigates the future network performance if an annual Status quo investment is made
	"Requested or Higher Budget" US\$ 1,500 million	Investigates the future network performance if a higher annual investment is made
	"Do Nothing" Budget	Analysed to determine the condition and deterioration of the network if no funding was available. No committed projects are analysed as part of this scenario.
"Technical Needs Budget"		Selects the first and earliest treatment that is triggered for each road segment, according to the intervention measures adopted for UNRA. This scenario is used to determine the funding level that is currently required to remove the backlog with regards to preventive maintenance, rehabilitation and upgrading to paved standards.

- Investment Plan

Situational Analysis

Needs Analysis (Maintenance, Rehabilitation and
Upgrading needs, Predicted
Condition, VOC, EUC, Asset Values)

Bridge Needs Analysis Report

Formed the basis of the Technical Criteria for NDPIII
Roads prioritization Process



- Upgrade of the ROMDAS System to IQL1
- HDM4 Calibration to Level 2
- 5 Frame work data collection contracts for 3 years
- Raodroid for Stations (Quarterly condition data)
- Automated Data collection Contracts at 30 Locations – *Frame work Contract for 3 Yrs.*
- Asset Management Support Specialist (Short term)
- Asset Management Support Contract – Consultancy (2yr)

- Skills not readily available on the Ugandan market
Accelerated on the job training
- Remuneration not inline with skill set required
Salary enhancement to match job skills
- Regular specialist training not easy to secure
Structured specialized training program
- Low exposure to changing advanced technology
Collaborative exchange programs with other roads authorities for hands on exposure
- Data collection Frequency requires more staff
Private consultants on frame work contracts

- Deliberate skilled staff retention through enhanced remuneration with binding contracts

Keep Technicians happy to be technicians

- Special Training and retraining programs
- Accelerated job exposure to technology with other institutions.



**THANK
YOU**

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